

AI Usage in CI/CD/CT Pipelines for Compute Platforms in Automotive - CARIAD Master Thesis

Overview

The primary aim of this thesis is to explore and implement AI-driven solutions within Continuous Integration (CI), Continuous Delivery (CD), and Continuous Testing (CT) pipelines, specifically focusing on automotive compute platforms.

This master thesis proposal outlines a research project focused on leveraging the power of Artificial Intelligence (AI) and Large Language Models (LLMs) to enhance the automation and effectiveness of security testing within CARIAD's development environment.

Research Objectives

- Investigate and develop novel security testing methods that integrate Whitebox and Blackbox fuzzing techniques with the capabilities of LLMs and AI.
- Define and implement the identified security testing into an existing or proposed Continuous Integration/Continuous Delivery/Continuous Testing (CI/CD/CT) pipeline.
- Develop mechanisms for the automatic generation of test cases, test procedures, test reports, and a product quality matrix based on the results of the implemented security testing methods.
- Explore and analyze techniques for the automatic generation of realistic and effective mock scenarios to enhance the scope and effectiveness of security testing.
- Analyze the impact of the implemented AI-driven security testing methods on key performance indicators such as test run time, efficiency in identifying vulnerabilities, and overall test coverage.

Expected Outcomes

- A well-researched and developed approach for integrating AI and LLMs into Whitebox and Blackbox fuzzing techniques for automotive software security testing.
- A practical implementation of these methods within a CI/CD/CT pipeline, demonstrating their feasibility and effectiveness.
- An automated system for generating comprehensive test artifacts (test cases, procedures, reports, quality matrix) based on the results of AI-driven fuzzing.
- Insights into the challenges and benefits of automatically generating mock scenarios for security testing in the automotive context.
- A quantitative analysis of the impact of these AI-driven techniques on the efficiency, coverage, and time effectiveness of security testing.

The findings of this thesis will contribute to the advancement of security testing practices in the automotive industry, specifically by exploring innovative ways to leverage AI and LLMs for enhanced automation and vulnerability detection. The practical implementation within CARIAD will provide valuable insights and potentially contribute to the improvement of their software development and security assurance processes.